

Emmanuel Katipunan

Impact-driven Software Developer & AI Engineer with expertise in designing full-stack software solutions and serverless architectures to solve complex business and community challenges. Proven track record in building AI-powered predictive analytics tools, automating high-volume workflows, and engineering high-availability systems using Nodejs, PHP, python, and cloud technologies. Passionate about leveraging AI and modern software design to optimize operations, enhance accessibility, and drive measurable organizational and societal outcomes.

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WORK EXPERIENCE

Callbox Inc.

Software Developer

June 2023 - August 2026

- Architected and deployed responsive full-stack web integrations and embedded micro-frontend applications using React, TypeScript, and Serverless Functions to consolidate cross-platform business workflows.
- Designed interactive custom web components and modal workflows to aggregate third-party SaaS data into unified views, eliminating application switching and boosting operational productivity.
- Scaled and optimized legacy software infrastructure to process high-volume API payloads and real-time webhook syncs, ensuring zero data loss and maintaining high availability under heavy traffic.
- Engineered AI-driven automation systems and background processing scripts to automate repetitive administrative workflows, significantly accelerating the timeline to meet core business quotas.
- Engineered an AI-powered, predictive reporting engine that aggregates complex company metrics to generate comprehensive performance and production forecasts, transforming raw data into highly digestible, actionable projections for leadership.
- Engineered an intelligent dialer tool integrated with AI/NLP capabilities that automatically analyzes call recordings, generates structured summaries, and extracts answers to key probing questions, reducing agent workflow time by 50%.

EDUCATION

West Visayas State University Calinog Campus, Calinog Iloilo

Bachelor of Science: Information And Communication Technology

June 2019 - March 2023

- Best Thesis awardee during the university research colloquium.
- Best Programmer awardee

PROJECTS

RaidDocs

July 2026

- Knowledge base system with retrieval augmented generation (RAG) powered by and optimized to 8b local LLM. (<https://github.com/emmsdev101/raid-docs>)

Yourlife

October 2021

- A social networking site with real-time notifications and real-time chat. It supports photo uploads and compresses it. Uses AWS bucket for object storage, MongoDB atlas for db and socket.io for real-time implementation. For security and authentication it uses passport and bcrypt. (<https://github.com/emmsdev101/YourLife>)

EchatMo

February 2021

- A web application where users can chat, create groups and community. It is a real-time chat application. (<https://github.com/emmsdev101/echatmo>)

Flappy-AI

September 2020

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A machine learning project that plays flappy bird autonomously. It has artificial neural network implementation in JavaScript. It uses reinforcement learning to evolve its weights or to learn (neuro evolution) (<https://github.com/emmsdev101/Flappy-AI>)

SKILLS

- Nodejs
- Python
- PHP
- Laravel
- Reactjs/Nextjs
- Relational Databases (mysql, postgresql)
- Nonrelational Databases (mongodb, redis, RavenDB ..etc)
- AI Agent orchestration
- Retrieval Augmented Generation